Cross3DReg: A Real-world Cross-source Point Cloud Registration Benchmark

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Abstract

Point cloud registration aligns point clouds acquired from different viewpoints or sensing devices. Cross-source point cloud registration specifically aims to align point clouds captured by heterogeneous LiDAR sensors with substantially divergent sampling patterns and densities. Compared to the same-source point cloud registration, cross-source registration faces two core challenges: the lack of publicly available real-world datasets for training the deep registration models and the inherent difference in point clouds captured by multiple sensors. The diverse patterns induced by sensors pose great challenges in robust and accurate point cloud feature extraction and matching, which negatively influence the registration accuracy. To address these challenges, we introduce Cross3DReg, a large-scale real-world outdoor benchmark collected using two heterogeneous LiDAR sensors: a rotating mechanical LiDAR and a hybrid semi-solid-state LiDAR. Moreover, an overlap-based cross-source registration framework is designed to utilize unaligned images to predict the overlapping region between the source and target. By filtering out redundant points in non-overlapping regions, the framework effectively reduces interference from noise and irrelevant points. Then, a visual-geometric

attention guided matching module is proposed to enhance the consistency of cross-source point cloud features by fusing image and geometric information to establish reliable correspondences and ultimately achieve accurate and robust registration. Extensive experiments show that our method achieves state-of-the-art registration performance. Our framework reduces the relative rotational error and relative translational error to **6.68°** and **1.01** m, respectively and increases the registration recall to **86.2%**, demonstrating its effectiveness in achieving cross-source point cloud registration.

Keywords: Cross-source point cloud registration, Hybrid semi-solid-state LiDAR, Rotating mechanical LiDAR, Multi-modal point cloud registration, Benchmark

1 Introduction

Cross-source point cloud registration [1, 2] aims to align point clouds acquired from heterogeneous sensors into a unified coordinate system. In modern 3D perception systems, individual sensors inevitably involve trade-offs among sensing range, spatial resolution, field of view and deployment cost. For example, mechanical LiDAR sensors typically provide long-range 360° coverage, but often produce relatively sparse and structured scanning patterns, whereas solid-state LiDAR sensors can generate denser measurements within a more limited field of view. To mitigate the limitations of a single sensing modality, autonomous systems increasingly rely on the fusion of complementary observations from diverse sensors. As a result, cross-source point cloud registration has become an important task for multi-sensor fusion [3], robust localization [4] and collaborative scene completion [5, 6].

Despite its importance, progress in cross-source registration has lagged behind that of same-source registration, primarily for two reasons. Firstly, publicly available benchmarks for this specific setting remain limited in number, scale and sensor realism. Existing datasets have made important contributions, but their sensing configurations and scales do not cover all aspects of real-world cross-source registration. 3DCSR [7] provides an early cross-source benchmark with indoor Kinect–LiDAR pairs, while KITTI-CrossSource [8] extends the evaluation to outdoor scenes using LiDAR and image-reconstructed point clouds. These datasets have enabled important progress in cross-source registration. However, they do not jointly provide a large-scale collection of outdoor point clouds captured by two heterogeneous LiDAR sensors. In particular, the image-reconstructed modality in KITTI-CrossSource does not contain all sensor-specific artifacts of a second physical LiDAR, such as its scanning pattern, outlier distribution and measurement noise. Consequently, the performance of existing methods on real cross-LiDAR data remains insufficiently characterized. Secondly, as shown in Figure 1, real cross-source point clouds exhibit significant discrepancies in data density, structural patterns, noise levels, outlier distributions and missing regions. While density variations also exist in single-source registration under varying viewpoints or partial observations [9], cross-source scenarios involve coupled differences combining density changes with sensor-specific scanning and measurement characteristics.

For instance, rotating LiDAR generates sparse ring-shaped point clouds, while semi-solid-state LiDAR produces fan-shaped point clouds. Such compound discrepancies are more challenging than pure density variations, often leading to extensive feature mismatches and impairing registration accuracy.

When applied to real-world cross-source point clouds, most current same-source registration methods [10–13] suffer performance drops. The main reason is feature inconsistency stemming from heterogeneous sensor differences in point density and structural patterns. In particular, for sparse cross-source point clouds with low overlap, models can hardly extract discriminative and consistent cross-source features, leading to numerous false correspondences and inaccurate pose estimation. Although several cross-source point cloud registration methods [1, 8, 14] are proposed, they are evaluated on synthetic datasets rather than real sensor-captured data. Consequently, their practical performance and limitations on the real cross-source point cloud data remain unexplored.

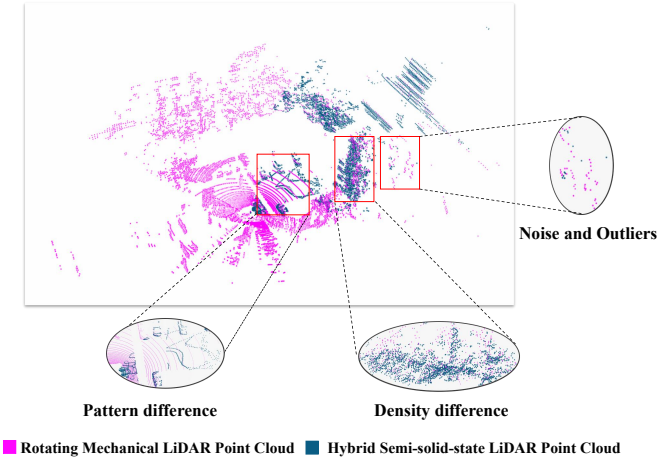


Fig. 1 Challenges of real-world cross-source point clouds, including differences in structural patterns, density and the presence of realistic noise.

To address the above challenges, we first construct a large-scale real-world cross-source point cloud registration dataset named Cross3DReg. The source point clouds are captured by a hybrid semi-solid-state LiDAR and the target point clouds are acquired with a 32-line rotating mechanical LiDAR. We also capture front views of the scenes from a viewpoint roughly co-located with the LiDAR. To mitigate interference from outliers and noise in point cloud matching, as well as the feature inconsistency arising from differences in cross-source point cloud density and structural patterns, we propose an overlap-based cross-source point cloud registration method by leveraging images to provide consistent visual clues. To this end, a dual-modal encoder is designed that effectively fuses image features with the coarse-grained geometric features of point clouds, which allows for predicting the probability that a point lies in

the overlap region between source and target point clouds. In this way, we can effectively filter out redundant points outside the image field of view. Having predicted the candidate points located in the overlap region between source and target point clouds, our goal is to establish correct correspondences among these points. However, cross-source point clouds exhibit different point patterns, distributions, resolutions and levels of noise and outliers, geometric-feature-based matching methods are less effective. Considering that correct correspondences within the overlap region should have consistent features and the common image information helps enhance the feature consistency, we propose a visual-geometric attention-guided matching module to enhance the feature consistency between cross-source point clouds by fusing visual and geometric information adaptively with an attention mechanism, thereby establishing reliable correspondences. Notably our method does not rely on calibration between cameras and LiDAR sensors. This is beneficial to avoid the inaccurate calibration caused by inevitable sensor vibration and the inherent extrinsic parameter drift. By leveraging images to enhance the consistency of point cloud features, our approach achieves accurate registration as long as the images are captured from a nearly co-located view, regardless of camera positions. This relaxed requirement for image input enhances the practical value of our approach. Experimental results also demonstrate the strong generalization capability of our method across images of varying quality and frames.

Our main contributions are listed below:

- Cross3DReg is a large-scale outdoor real-world cross-source point cloud registration dataset. It includes 13,231 point cloud pairs where different levels of noise, outliers, densities and structural patterns are present. Images showing common views between source and target point clouds are also collected.
- An overlap-based cross-source point cloud registration method is proposed to achieve accurate registration by predicting the overlap region with the help of unaligned images and ignoring redundant points and noise.
- To achieve accurate feature matching within the overlap region, a visual-geometric attention-guided matching module is proposed to fuse visual and geometric information adaptively with an attention mechanism, enhancing feature consistency of points within the overlap region between cross-source point clouds.

2 Related Work

Same-source point cloud registration. Currently, same-source point cloud registration methods can be categorized into three classes. The first class is traditional iterative optimization-based approaches [15–17]. However, these methods suffer from a significant drop in registration accuracy when dealing with noisy or structurally complex scenes. The second class is correspondence-based point cloud registration methods [10, 18, 19]. Early approaches primarily rely on handcrafted feature descriptors [20, 21] to establish point-wise correspondences. With the rapid development of deep learning in point cloud registration, a series of deep learning-based feature extraction models [22, 23] have been proposed, enabling more accurate and robust registration. Nevertheless, these methods still experience notable performance degradation when applied to regions with low overlap or a large number of outliers. To address these challenges,

coarse-to-fine registration strategies [11, 24] have recently been adopted, showing accurate registration results under conditions of low overlap and high noise levels. The third class is end-to-end point cloud registration methods [25–27]. Unlike conventional two-stage registration frameworks, end-to-end approaches directly utilise deep neural networks to predict rigid transformations between point clouds without explicitly establishing point correspondences, thereby improving overall registration efficiency. Moreover, since raw point cloud data contains only geometric information, recent multimodal point cloud registration methods [28–30] attempt to enhance feature discriminability by incorporating additional modalities (like colour, semantics, texture), allowing for more reliable correspondences. However, they all rely on explicit camera calibration to achieve geometric correspondence between pixel and point cloud, making them difficult to adapt to real-world scenarios.

Cross-source point cloud registration. The core challenge in cross-source point cloud registration lies in addressing the significant discrepancies introduced by different types of sensors. Compared to the same-source registration, cross-source point clouds exhibit a gap in density and pattern distribution and are more susceptible to outliers. Traditional methods [31, 32] are not designed to cope with these problems. To address the specific challenges posed by cross-source data, in recent years, mainstream cross-source point cloud registration methods [1, 8, 33] widely adopt a coarse-to-fine strategy. Correspondences are established by learning consistent deep features between the cross-source point clouds, thereby achieving robust registration. However, the advancement of this field is constrained by limited datasets; the efficacy of current approaches [1, 14] has primarily been validated on cross-source datasets where the source/target point clouds are synthesised with images, not real-world cross-source datasets. Therefore, developing cross-source point cloud registration data using real-world sensors represents a key future research objective in this area.

3 The Cross3DReg Dataset

To facilitate the development of cross-source point cloud registration, we introduce a real-world Cross3DReg dataset that contains $13,231$ point cloud pairs captured by a rotating mechanical LiDAR and a hybrid semi-solid-state LiDAR. The images are collected at a co-located view with the hybrid semi-solid-state LiDAR. And only one sensor is activated during a single data collection session. Table 1 shows the differences of the Cross3DReg compared with point cloud registration datasets. Cross3DReg further incorporates image modality information and contains sparse point clouds collected in outdoor scenes by two real sensors with different scanning patterns, which are typically absent from same-source datasets. Beyond the basic dataset statistics, we further analyze these datasets from the perspectives of spatial point distribution and registration difficulty.

Here, we present the devices we used to collect the dataset, the scanning process and the overlap information between the source and target point clouds of Cross3DReg dataset in the following.

The capturing equipment. As shown in Figure 2, our Cross3DReg dataset is collected by a custom-built Unmanned Ground Vehicle, equipped with a 32-beam

Table 1 The comparison of point cloud registration datasets. Po: Partial overlap. Rno: Real-world noise and outliers. Rdp: Real difference of structural patterns. Dens: Density difference. Open: Open source. RL: Rotating mechanical LiDAR. HL: Hybrid semi-solid-state LiDAR.

Type	Dataset	Scenes	Sensors	Pairs	Img.	Po.	Rno.	Rdp.	Dens.	Open.
same-source	3DMatch [34]	indoors	RGB-D	25377	✓	✓	×	×	×	✓
	KITTI [35]	outdoors	LiDAR	1780	✓	×	✓	×	×	✓
cross-source	3DCSR [2]	indoors	Kinect, LiDAR, SFM	202	×	✓	×	×	✓	✓
	KITTI-CrossSource [8]	outdoors	LiDAR, MonoRec	2006	×	✓	×	×	✓	✓
	Cross3DReg	outdoors	RL, HL	13231	✓	✓	✓	✓	✓	✓

rotating mechanical LiDAR, a hybrid semi-solid-state LiDAR, an RGB-D stereo camera (ZED camera) and a RGB camera. Due to the high quality of captured images, in our dataset, RGB images acquired with the left camera of ZED are used as the visual information and available to the public. And the reasons for employing these sensors can be summarized as follows:

- According to our research, the primary representative cross-source point cloud datasets currently available are 3DCSR [7] and KITTI-CrossSource [8]. 3DCSR comprises 202 pairs of indoor LiDAR and SFM cross-source point clouds, whilst KITTI-CrossSource contains 2006 pairs of outdoor LiDAR and MonoRec cross-source point clouds. It should be noted that at least one point cloud in these datasets is synthesised; the academic community currently lacks cross-source point cloud data entirely derived from real sensor acquisitions.
- From a sensor characteristics perspective, the rotating mechanical LiDAR can provide high-precision, 360° omnidirectional scans, making it a staple for high-fidelity mapping. In contrast, the hybrid semi-solid-state LiDAR offers a more cost-effective solution with a constrained field of view, typically found in mass-produced autonomous mobile platforms. These two modalities exhibit significant discrepancies in scanning patterns, point densities and noise characteristics, which fully reveal the core challenges inherent in cross-source point cloud registration. Additionally, our dataset can also be used in heterogeneous LiDAR place recognition [36], localization [37] for an embodied agent in diverse degenerate scenarios.

The scanning process. We design and execute 11 distinct collection routes across a university campus. These routes encompass typical campus environments, featuring a rich variety of dynamic and static elements, which include dynamic traffic participants (e.g., vehicles and pedestrians), static obstacles (e.g., road barriers), as well as infrastructure and natural landscapes (e.g., buildings and vegetation). Furthermore, to preserve modality independence under authentic cross-source scenarios, only a single sensor is activated per data collection session. Notably, the ZED camera and hybrid semi-solid-state LiDAR are configured with largely congruent fields of view, ensuring sufficient spatial overlap for registration tasks.

Dataset information. We split the dataset into training, validation and test sets. Training data is drawn from routes 00-05 ($7,772$ pairs), validation data from routes 06–07 ($1,314$ pairs) and test data from routes 08–10 ($4,145$ pairs). For every source – target point cloud pair, the ground-truth is the rigid transformation that precisely registers the source to the target. For the generation of the ground-truth transformation matrices, we randomly sample the rotation angle and

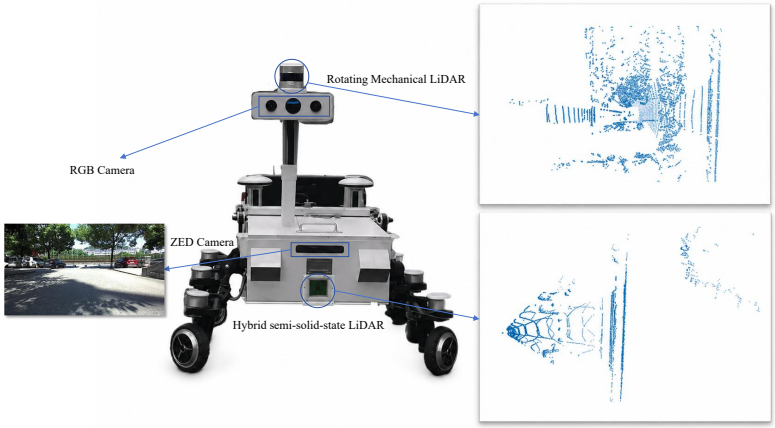


Fig. 2 Data acquisition platform. We adopt two heterogeneous LiDAR sensors for point cloud collection, namely the rotating mechanical LiDAR (Ouster OS1-32) and the hybrid semi-solid-state LiDAR (Livox). The Ouster OS1-32 is a 32-channel rotating LiDAR sensor that delivers a 360° horizontal field of view (FoV) with a maximum detection range of approximately 120 m. In comparison, the Livox sensor supports a detection range of 0–260 m, which is reduced to 0–90 m under 10% reflectivity, along with a horizontal FoV of 81.7° and a vertical FoV of 25.1°. Benefiting from its hybrid semi-solid-state structure, Livox adopts a non-repetitive scanning strategy, where laser beams are distributed within angular region centered on the forward optical axis. Different from the Ouster OS1-32, which provides full 360° horizontal coverage, the Livox sensor only captures point cloud measurements within its forward-facing field of view. RGB images are acquired using the left camera of a ZED stereo camera, whose viewpoint is spatially aligned with the Livox sensor. All sensors are configured to operate at a frequency of 10 Hz, ensuring consistent temporal synchronization among heterogeneous data streams.

the translation magnitude. Specifically, the rotation angle is sampled within the range of 5° to 180°, while the translation magnitude is sampled from 3 to 15 m. This setting enables the dataset to cover a wide range of challenging pose variations, including large-rotation/large-translation, small-rotation/small-translation, large-rotation/small-translation and small-rotation/large-translation cases. Given two point clouds $\mathcal{P}_{src}$, $\mathcal{Q}_{ref}$, we compute the overlap ratio between two point clouds as follows:

$$\mathcal{P}^{'}_{src} = Trans(\mathcal{P}_{src}), \tag{1}$$

$$\mathcal{O}_{ref \to src} = \frac{1}{|\mathcal{Q}_{ref}|} \sum_{i=1}^{|\mathcal{Q}_{ref}|} \mathbb{I}(min||q_i - p'||_2 < r), \tag{2}$$

$$\mathcal{O}_{src \to ref} = \frac{1}{|\mathcal{P}^{'}_{src}|} \sum_{j=1}^{|\mathcal{P}^{'}_{src}|} \mathbb{I}(min||p_j - q||_2 < r), \tag{3}$$

$$\mathcal{O} = min(\mathcal{O}_{ref \to src}, \mathcal{O}_{src \to ref}), \tag{4}$$

where $\mathcal{Q}^{'}_{ref} = \{q_i \in \mathbb{R}^3 | i = 1, ..., m\}$; $\mathcal{P}_{src} = \{p_j \in \mathbb{R}^3 | j = 1, ..., n\}$; $Trans(.)$ represents the rigid transformation; $||.||_2$ is the Euclidean distance; $\mathbb{I}(.)$ is the indicator

function; r denotes the distance threshold. Based on a calculated average inter-point distance of 0.2m, we set the distance threshold r to 0.2m. Figure 4 (g), (h), (i) respectively present the statistics of rotation magnitude, translation magnitude and overlap ratio in the Cross3DReg dataset. The transformation space in our dataset thoroughly spans various challenging configurations, ranging from minor adjustments to extreme cases of substantial rotation and translation. Moreover, the inherent difficulty is compounded by the sparsity of overlapping regions, with the average overlap ratio maintained below 0.3.

Comparison with existing cross-source datasets. To further characterize the uniqueness and difficulty of Cross3DReg, we compare it with existing cross-source point cloud datasets from two perspectives: spatial point distribution and dataset-level registration properties, including ground-truth transformation ranges and overlap ratios. As shown in Figures 3 (a) and (b), 3DCSR mainly consists of indoor scene data, where both LiDAR and Kinect point clouds exhibit sharp and concentrated density peaks in the near-range region of approximately 2-3m. However, there are still significant differences in the spatial distribution of the two. LiDAR offers superior long-range measurement capabilities and can cover a larger spatial area, but its point cloud typically exhibits local sparsity and scanning discontinuities; in contrast, Kinect point clouds are primarily concentrated in the near field, with denser local sampling and a relatively more uniform overall distribution. Moreover, as shown in Figure 4 (a), (b) and (c), the rotation range of 3DCSR is approximately 0°-135° and the translation range is 0-0.6m. Meanwhile, the overlap ratios of source and target point clouds are mainly concentrated between 0.6 and 1.0, indicating that the dataset consists mainly of high-overlap registration scenarios.

For KITTI-CrossSource, the two types of point clouds show more pronounced differences in their probability density distributions along the distance dimension. As presented in Figure 3 (c) and (d), LiDAR point clouds in the KITTI-CrossSource dataset exhibit typical distance-dependent sampling characteristics, where a large number of points are concentrated in the near-to-middle range and the point density gradually decreases as the distance increases. In contrast, the synthesized point clouds present distinctly different distance distribution patterns, characterized by more prominent density peaks in far-range regions, higher point densities within certain distance intervals, weakened scanline structures and more continuous spatial distributions. As shown in Figure 4 (a)-(c), rotations in KITTI-CrossSource are mainly concentrated within the range of 0°–1°, while the translation range is approximately 6–10m. The overlap ratios are generally lower than 0.4 and are mainly concentrated between 0.1 and 0.3.

Compared with 3DCSR and KITTI-CrossSource, as shown in Figure 3 (e)-(f) and Figure 4 (g)-(i), our Cross3DReg dataset provides a more diverse and challenging distribution of ground-truth transformations. Specifically, the rotation angles cover a range of 5°–180° and the translation magnitudes cover a range of 3–15m. Meanwhile, Cross3DReg exhibits lower overall overlap ratios, with many samples having overlap ratios below 0.3, demonstrating significant low-overlap characteristics. Therefore, Cross3DReg can serve as a challenging and complementary benchmark dataset

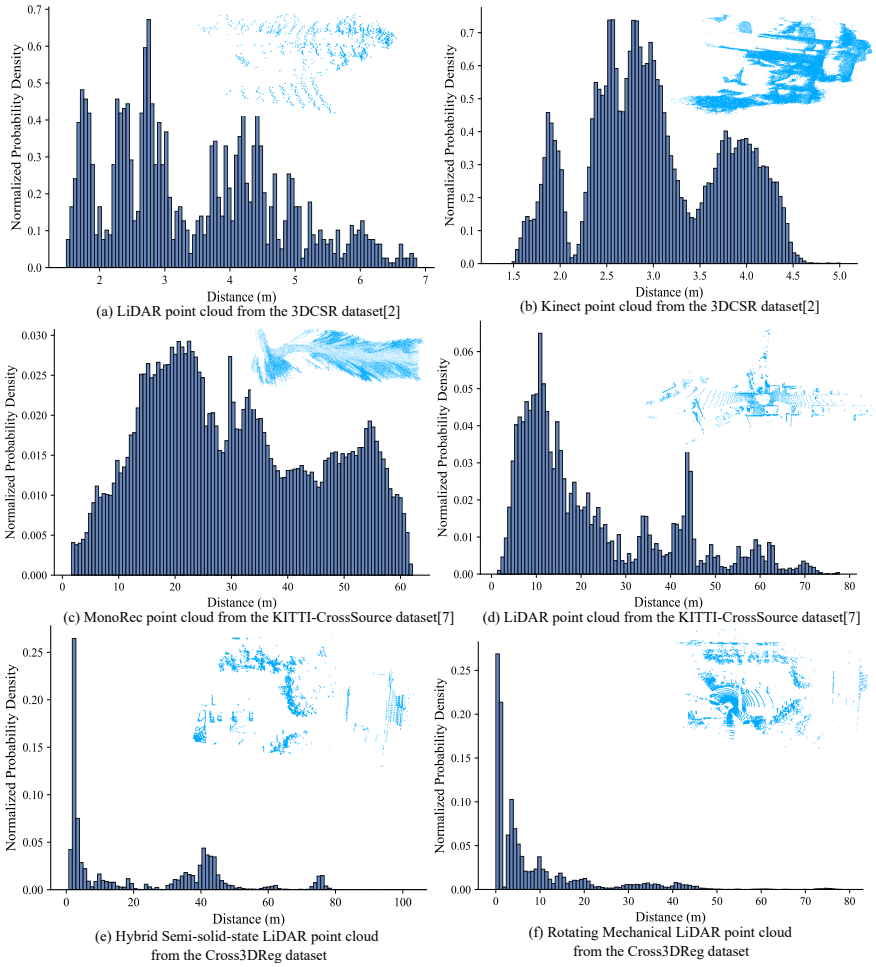


Fig. 3 Comparison of point-to-sensor distance distributions and representative point-cloud structures across cross-source datasets. Each panel shows the normalized histogram of point-to-sensor distance together with a representative scan. Panels (a)–(b) correspond to the LiDAR and Kinect point clouds in 3DCSR; panels (c)–(d) correspond to the MonoRec and LiDAR point clouds in KITTI-CrossSource; panels (e)–(f) correspond to the hybrid semi-solid-state LiDAR and rotating mechanical LiDAR point clouds in Cross3DReg. The two physical LiDARs in Cross3DReg exhibit different and non-uniform spatial distributions, the sensor-specific density and scanning-pattern discrepancies addressed in this work.

for cross-source point cloud registration, particularly for evaluating registration methods under large pose variations, heterogeneous sensor distribution discrepancies and low-overlap scenarios.

Examples of the Cross3DReg dataset are visualized in Figure 14. Source and target point clouds are shown in the first two columns. It is obvious to see that the Cross3DReg dataset is challenging where different point patterns, level of noise and

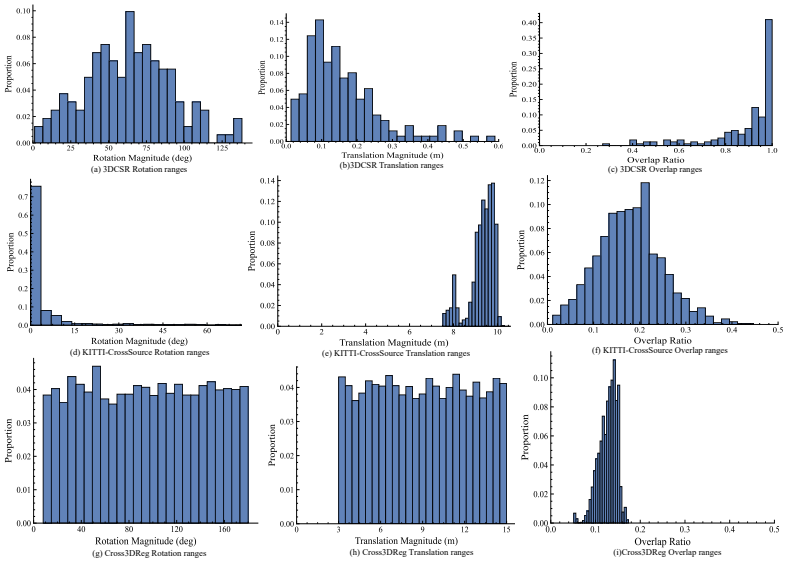


Fig. 4 Distributions of ground-truth relative pose and overlap ratios in 3DCSR, KITTI-CrossSource and Cross3DReg. The columns show rotation magnitude (degrees), translation magnitude (meters) and overlap ratio, respectively; the rows correspond to 3DCSR, KITTI-CrossSource and Cross3DReg. Compared with 3DCSR, which mainly contains high-overlap pairs and KITTI-CrossSource, which has mostly small rotations and overlap ratios concentrated around 0.1–0.3, Cross3DReg covers a wider pose range of approximately 5°–180° and 3–15 m, while most pairs have overlap ratios below 0.3.

outliers and various densities are present. Corresponding images and aligned point clouds are also presented in the last two columns.

4 Method

Problem Statement. The cross-source point cloud registration is formulated as follows. Given the source point cloud $\mathcal{P} = \{\mathbf{p}_i \in \mathbb{R}^3 | i = 1, \ldots, N\}$ and target point cloud $\mathcal{Q} = \{\mathbf{q}_j \in \mathbb{R}^3 | j = 1, \ldots, M\}$, we estimate an optimal rigid transformation matrix $\mathcal{T} = \{\mathbf{R}, \mathbf{t}\}$ to align $\mathcal{P}$ and $\mathcal{Q}$, where $\mathbf{R} \in SO(3)$ is the rotation matrix and $\mathbf{t} \in \mathbb{R}^3$ is the translation vector. The optimisation goal can be formulated as:

$$\mathcal{T} = \arg\min_{R,t} \sum_{(p_i, q_j) \in C^*} \|\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_j\|_2, \tag{5}$$

where $\mathcal{C}^*$ represents the correspondences between $\mathcal{P}$ and $\mathcal{Q}$ and the $\|.\|_2$ denotes the Euclidean distance.

For accurate cross-source registration, we propose an overlap-based cross-source point cloud registration method to establish the correspondence between point clouds. As shown in Figure 5, the framework contains four phases: 1) Feature extraction. Features of Image $\mathcal{I}$ and point clouds $\mathcal{P}$ and $\mathcal{Q}$ are extracted through a two-branch network. The point cloud branch downsamples the original point clouds to acquire

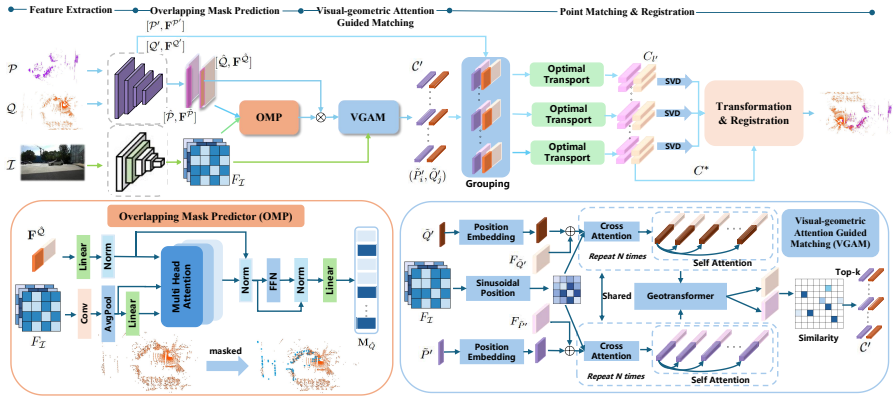


Fig. 5 Our Cross3DReg registration method consists of four parts. We first encode the source and target point clouds, $\mathcal{P}$ and $\mathcal{Q}$ and an unaligned image $\mathcal{I}$, to extract superpoints and corresponding features. Based on the OMP module, image features are then fused with each point cloud's features to predict if superpoints are within the image view with the predicted binary mask vector $\mathbf{M}_{\hat{Q}}$ for the target (and $\mathbf{M}_{\hat{P}}$ for the source). With these superpoint sets of the source and target, the VGAM module establishes superpoint correspondences $\mathcal{C}'$ between source and target point clouds based on the visually enhanced features. Finally, these established correspondences are propagated to the original dense point clouds to generate final correspondences $\mathcal{C}_{l'}$, which are fed into a pose estimator to compute the transformation matrix.

superpoints $\hat{\mathcal{P}}$ and $\hat{\mathcal{Q}}$. The corresponding features $\mathbf{F}^{\hat{\mathcal{P}}}$ and $\mathbf{F}^{\hat{\mathcal{Q}}}$ are also extracted; 2) Overlapping Mask Predictor (OMP). Although images are not aligned with point clouds, they contain the common views of the source and target point clouds. Thus, the coarse-grained superpoint features are firstly fused with image features to predict the mask of superpoints that indicates if the superpoint is within the image view. Both the overlap masks of the source and target point clouds are predicted. The following matching step is only performed between overlapping masks. 3) Visual-Geometric Attention guided superpoint Matching (VGAM). With the overlapping masks, visually enhanced superpoint features are utilised for similarity computation and robust coarse-grained correspondences are established between the source and target. 4) Point matching and registration. Based on the superpoint matching results, precise point-level correspondences are obtained by local point cloud feature matching. Based on the point correspondences, transformations are estimated using the pose estimators.

4.1 Feature Extraction

Due to the significant density variations and a large number of points, we first preprocess the raw point clouds using a voxel-based downsampling method (voxel size = 0.25) before feeding them into the feature extractor. The downsampled source point cloud $\mathcal{P}$ and the target point cloud $\mathcal{Q}$ are then put into the KPConv-FPN backbone network [38], which can extract the point cloud features at different scales. At the coarsest scale, we obtain the superpoint sets $\hat{\mathcal{P}}$ and $\hat{\mathcal{Q}}$. Their corresponding features are $\mathbf{F}^{\hat{\mathcal{P}}} \in \mathbb{R}^{|\hat{P}| \times \hat{d}}$ and $\mathbf{F}^{\hat{\mathcal{Q}}} \in \mathbb{R}^{|\hat{Q}| \times \hat{d}}$, $\hat{d}$ is the feature dimension of superpoints at the

coarsest level. Points at the densest level are denoted as $\mathcal{P}', \mathcal{Q}'$ and their corresponding features are $\mathbf{F}^{\mathcal{P}'} \in \mathbb{R}^{|\mathcal{P}'| \times d'}$ and $\mathbf{F}^{\mathcal{Q}'} \in \mathbb{R}^{|\mathcal{Q}'| \times d'}$, d' is the corresponding feature dimension.

For image feature extraction, we employ a U-Net backbone network with residual connections to process the intermediate unaligned images. Given an input image $\mathcal{I} \in \mathbb{R}^{H \times W}$, its feature is represented as $F_{\mathcal{I}} \in \mathbb{R}^{H \times W \times d}$.

4.2 Overlapping Mask Predictor Module

As various point densities, prevalent noise, outliers and different structural patterns exist in the cross-source point cloud registration, traditional point features-based overlapping region estimation is inaccurate. Since the captured images in our dataset contain the common views, we can utilise these images to identify possible points that are located in the overlap region between the source and target point clouds. This is also beneficial to filter redundant points out of the image view.

Therefore, we propose an overlapping mask prediction module based on misaligned images. We define this problem as a binary regression task, fusing image information to directly perform linear regression for the mask of each superpoint. Since the camera calibration information is unavailable, images and point clouds are unaligned. We first perform image and point cloud feature dimension alignment. The image features $\mathbf{F}_{\mathcal{I}} \in \mathbb{R}^{H \times W \times d}$ and superpoint features $\mathbf{F}^{\hat{\mathcal{Q}}} \in \mathbb{R}^{|\hat{Q}| \times \hat{d}}$ are mapped to a unified dimensional space by linear projection to obtain aligned features. The aligned image features and the superpoint features of point clouds are represented as $\widehat{\mathbf{F}}^{\hat{\mathcal{Q}}}$ and $\widehat{\mathbf{F}}_{\mathcal{I}}$. Then, we use the multi-attention mechanism [39] to carry out the cross-modal feature fusion. Finally, the fused features are processed by a residual feed-forward network (FFN) with layer normalization. The overlap probability of superpoints is output via an MLP net. Taking the prediction of the overlap mask for the target point cloud $\mathbf{Q}$ as an example, the process can be formalized as follows.

$$\mathbf{F}_{fuse} = MultiHeadAttn(\widehat{\mathbf{F}}^{\hat{\mathcal{Q}}}, \widehat{\mathbf{F}}_{\mathcal{I}}). \tag{6}$$

Given the fused image and point cloud features, the probability of each superpoint belonging to the overlap region, denoted as $\mathbf{P}^{\hat{Q}}_{overlap}$, can be estimated as:

$$\mathbf{P}^{\hat{Q}}_{overlap} = \sigma(MLP(\mathbf{F}_{fuse} + (FFN(\mathbf{F}_{fuse} + \widehat{F}^{\hat{\mathcal{Q}}})))), \tag{7}$$

$$\mathbf{M}_{\hat{Q}_i} = \begin{cases} 1, & \text{if } \mathbf{P}^{\hat{Q}_i}_{\text{overlap}} > \lambda, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

where $\mathbf{M}_{\hat{Q}} \in \{0, 1\}$ is a binary mask vector; σ represents the sigmoid function; and λ denotes the confidence threshold with a default of 0.5.

4.3 Visual-Geometric Attention Guided Matching

To effectively incorporate visual context into geometric features for point cloud registration, we introduce the visual-geometric attention guided matching module to

leverage both visual and geometric information to enhance the consistency of superpoint features between point cloud of different sources. The core of our approach is a two-stage attention mechanism. First, the visual cross-attention mechanism fuses visual context from image features into the superpoint features. Subsequently, a geometric self-attention mechanism [24] refines these fused features to capture global geometric relationships within the point cloud.

Let's take the target point cloud $\mathcal{Q}$ as an example. We first use a predicted overlap mask $\mathbf{M}_{\tilde{Q}}$ to select a subset of superpoints $\tilde{\mathbf{Q}}'$ and their corresponding features $\mathbf{F}_{\tilde{Q}'}$. The corresponding image features $\mathbf{F}_I$ are flattened into a vector. To provide the attention mechanism with spatial awareness, we introduce positional encoding, $\mathbf{F}_{pos}^{\tilde{Q}'}$ for super points and $\mathbf{F}_{pos}^{I}$ for image pixels.

The superpoint features, image features and their respective positional encodings are projected into Query ($\mathbf{Q}_c$), Key ($\mathbf{K}_c$)and Value ($\mathbf{V}_c$) spaces using learnable linear matrices. The positional encodings are also projected to generate point cloud positional embeddings ($\mathbf{E}_c$) and image positional embeddings ($\mathbf{G}_c$). The process is formulated as:

$$\mathbf{Q}_c = \mathbf{F}_{\tilde{Q}'}\mathbf{W}_{Q_c}, \quad \mathbf{K}_c = \mathbf{F}_I\mathbf{W}_{K_c}, \quad \mathbf{V}_c = \mathbf{F}_I\mathbf{W}_{V_c}, \tag{9}$$

$$\mathbf{E}_c = \mathbf{F}_{pos}^{\tilde{Q}'}\mathbf{W}_{E_c}, \quad \mathbf{G}_c = \mathbf{F}_{pos}^{I}\mathbf{W}_{G_c}, \tag{10}$$

where $\mathbf{W}_{Q_c}, \mathbf{W}_{K_c}, \mathbf{W}_{V_c}$ are learnable projection matrices for the Query, Key and Value, respectively. And $\mathbf{W}_{E_c}, \mathbf{W}_{G_c}$ are the projection matrices for their respective positional embeddings.

By integrating content and position information, we compute the cross-attention scores. These scores weigh the aggregation of the Value and image positional embeddings. The superpoint features are then updated via a residual connection:

$$\mathbf{Scores}_c = \mathit{softmax}\left(\frac{(\mathbf{Q}_c + \mathbf{E}_c)(\mathbf{K}_c + \mathbf{G}_c)^T}{\sqrt{d'}}\right), \tag{11}$$

$$\mathbf{F}'_{\tilde{Q}'} = \mathrm{Scores}_c(\mathbf{V}_c + \mathbf{G}_c) + \mathbf{F}_{\tilde{Q}'}, \tag{12}$$

where $\mathbf{F}'_{\tilde{Q}'}$ denotes the updated superpoint features enriched with visual context information.

To enhance the global structural integrity of the features and mitigate potential noise, we employ a self-attention mechanism. This step promotes information propagation across entire point clouds. The visually-enhanced features $\mathbf{F}'_{\tilde{Q}'}$ are linearly projected into a new set of Query ($\mathbf{Q}_s$), Key ($\mathbf{K}_s$) and Value ($\mathbf{V}_s$). Self-attention weights are then computed to update the features in a residual manner:

$$\mathbf{Scores}_s = \mathit{softmax}\left(\frac{\mathbf{Q}_s\mathbf{K}_s^T}{\sqrt{d'}}\right), \tag{13}$$

$$\mathbf{F}''_{\tilde{Q}'} = \mathbf{Scores}_s\mathbf{V}_s + \mathbf{F}'_{\tilde{Q}'}. \tag{14}$$

Finally, combined with the geometric self-attention mechanism, we can maximize the descriptive power of the final features. After the feature enhancement process, we can obtain highly discriminative superpoint features, $\bar{\mathbf{F}}_{\tilde{P}'}$ and $\bar{\mathbf{F}}_{\tilde{Q}'}$, for the source and target point clouds, respectively. We then construct a feature similarity matrix $\mathbf{Z}'$ by the following formulation:

$$\mathbf{Z}'_{ij} = \exp\left(-\|\bar{\mathbf{F}}_{\tilde{P}'_i} - \bar{\mathbf{F}}_{\tilde{Q}'_j}\|_2^2\right), \tag{15}$$

where $\mathbf{Z}'_{ij}$ measures the similarity between the i-th source superpoint $\tilde{\mathbf{P}}'_i$ and the j-th target superpoint $\tilde{\mathbf{Q}}'_j$. Finally, we apply dual normalization [40, 41] to the similarity matrix $\mathbf{Z}'$ and select the top-K entries with the highest scores to form the final set of superpoint correspondences $\mathcal{C}' = \{(\tilde{\mathbf{P}}'_i, \tilde{\mathbf{Q}}'_j) | \tilde{\mathbf{P}}'_i \in \hat{\mathcal{P}}, \tilde{\mathbf{Q}}'_j \in \hat{\mathcal{Q}}\}$.

4.4 Point Matching and Registration

After obtaining the superpoint correspondences in the overlapping regions, we employ a point-to-node grouping strategy [11] to further establish correspondences between the dense points. The core idea of this strategy is to assign dense points to their nearest neighboring superpoints based on spatial distance. Specifically, for a matched superpoint pair $(\tilde{\mathbf{P}}'_i, \tilde{\mathbf{Q}}'_j)$, we denote their corresponding dense point groups as $\mathbf{G}_{\tilde{P}'_i}$ and $\mathbf{G}_{\tilde{Q}'_j}$ and their feature groups as $\mathbf{G}_i^{\mathbf{F}^{\mathcal{P}'}}$ and $\mathbf{G}_j^{\mathbf{F}^{\mathcal{Q}'}}$, respectively. Based on the superpoint correspondence $(\tilde{\mathbf{P}}'_i, \tilde{\mathbf{Q}}'_j)$, we compute the similarity matrix between the feature groups $\mathbf{G}_i^{\mathbf{F}^{\mathcal{P}'}}$ and $\mathbf{G}_j^{\mathbf{F}^{\mathcal{Q}'}}$ as $\mathbf{S}_{l'} = \frac{\mathbf{G}_i^{\mathcal{P}'}(\mathbf{G}_j^{\mathbf{F}^{\mathcal{Q}'}})^T}{\tilde{d}}$, where $\tilde{d}$ denotes the feature dimension. To enhance matching robustness, we adopt the method from [42] by adding slack terms, controlled by a learnable parameter α, to the last row and column of the similarity matrix $\hat{\mathbf{S}}'$. Subsequently, we apply the Sinkhorn algorithm to find the optimal matching. After removing the slack terms, we select the top K' matching pairs with the highest confidence scores to establish the group-level dense point correspondences $C_{l'}$. Finally, by aggregating the correspondences from all groups, we obtain the global correspondences $\mathcal{C}^* = \bigcup_{l'=1}^{|C'|} \mathcal{C}_{l'}$. Based on the correspondences, we employ the LGR estimator [24] to accurately estimate the transformation matrix.

4.5 Loss Function

Our loss function is composed of three components and can be expressed as: $\mathcal{L}_{total} = \mathcal{L}_{coarse} + \mathcal{L}_{fine} + \mathcal{L}_{mask}$. Here, following the framework of GeoTrans [24], $\mathcal{L}_{coarse}$ and $\mathcal{L}_{fine}$ are the losses supervising the coarse-grained (superpoint) and fine-grained (point-level) matching, respectively. For L_{mask}, we employ the Focal Loss [43]. Due to the lack of camera intrinsic and extrinsic parameters, which prevents the establishment of a projection relationship from the 3D point cloud to the 2D image, we adopt a mask generation strategy based on the point cloud overlap. Specifically, for a given source point cloud $\mathcal{P}$ and a target point cloud $\mathcal{Q}$, the ground-truth overlapping mask at the

superpoint level, $\mathbf{M}_g^i$, is defined as:

$$\mathbf{M}_g^i = \begin{cases} 1 & \text{if } \hat{\mathcal{P}}_i \text{ correspondent to } \hat{\mathcal{P}}_j \\ 0 & \text{otherwise} \end{cases}, \tag{16}$$

$$\mathbf{p}_t^i = \begin{cases} p_i & \text{if} \quad \mathbf{M}_g^i = 1 \\ 1 - p_i & \text{otherwise} \end{cases}, \tag{17}$$

$$\mathcal{L}_{mask} = \frac{1}{|\mathbf{M}_g|} \sum_{i}^{|\mathbf{M}_g|} -\alpha(1 - \mathbf{p}_t^i)^\gamma \log(\mathbf{p}_t^i), \tag{18}$$

where $\mathbf{p}_i$ denotes the mask probability of model output. $\mathbf{p}_t^i$ denotes the probability that the mask value is true. $\gamma = 2.0$ and $\alpha = 0.25$ are the focusing parameter and balancing parameter, respectively.

4.6 Pseudocode of Cross3DReg

Algorithm 1 presents the pseudocode of the proposed Cross3DReg method, detailing its overall process.

Algorithm 1 Cross3DReg Approach

Input: source points $\mathcal{P}$, target points $\mathcal{Q}$, unaligned image $\mathcal{I}$.
Output: Transformation matrix $\mathcal{T}$

$(\hat{\mathcal{P}}, \mathcal{P}'), (\mathcal{F}_{\hat{\mathcal{P}}}, \mathcal{F}_{\mathcal{P}'}) \leftarrow$ PointFeatureExtractor($\mathcal{P}$)
$(\hat{\mathcal{Q}}, \mathcal{Q}'), (\mathcal{F}_{\hat{\mathcal{Q}}}, \mathcal{F}_{\mathcal{Q}'}) \leftarrow$ PointFeatureExtractor($\mathcal{Q}$)
$\mathcal{F}^{\mathcal{I}} \leftarrow$ ImgFeatureExtractor($\mathcal{I}$)
$\mathbf{P}^{\hat{Q}}_{overlap}, \mathbf{P}^{\hat{P}}_{overlap} \leftarrow$ OMP($\hat{\mathcal{P}}, \mathcal{F}_{\hat{\mathcal{P}}}, \hat{\mathcal{Q}}, \mathcal{F}_{\hat{\mathcal{Q}}}, \mathcal{F}^{\mathcal{I}}$)
$\mathbf{M}_{\hat{\mathcal{P}}} \leftarrow (\mathbf{P}^{\hat{P}}_{overlap} > \lambda)$, $\mathbf{M}_{\hat{\mathcal{Q}}} \leftarrow (\mathbf{P}^{\hat{Q}}_{overlap} > \lambda)$
$\bar{F}_{\tilde{P}'_i}, \bar{F}_{\tilde{Q}'_j} \leftarrow$ VGAM($\hat{\mathcal{P}}, \mathcal{F}_{\hat{\mathcal{P}}}, \mathbf{M}_{\hat{\mathcal{P}}}, \hat{\mathcal{Q}}, \mathcal{F}_{\hat{\mathcal{Q}}}, \mathbf{M}_{\hat{\mathcal{Q}}}$)
for $i \leftarrow 1$ to N **do**
  for $j \leftarrow 1$ to M **do**
    $Z'_{ij} \leftarrow \exp\left(-\|\bar{F}_{\tilde{P}'_i} - \bar{F}_{\tilde{Q}'_j}\|_2^2\right)$
  end for
end for
$\mathcal{C}' \in \{(\tilde{\mathbf{P}}'_i, \tilde{\mathbf{Q}}'_j) | \tilde{\mathbf{P}}'_i \in \hat{\mathcal{P}}, \tilde{\mathbf{Q}}'_j \in \hat{\mathcal{Q}}\} \leftarrow$ select top K Z'_{ij}
$\mathbf{G}_{\tilde{\mathbf{P}}'}, \mathbf{G}_{\tilde{\mathbf{Q}}'} \leftarrow GroupPoints(\mathcal{C}', \mathcal{Q}', \mathcal{P}')$
$\mathbf{G}^{\mathbf{F}^{\mathcal{P}'}}, \mathbf{G}^{\mathbf{F}^{\mathcal{Q}'}} \leftarrow GroupFeats(\mathbf{G}_{\tilde{\mathbf{Q}}'}, \mathbf{G}_{\tilde{\mathbf{P}}'} \mathcal{F}_{\mathcal{Q}'}, \mathcal{F}_{\mathcal{P}'})$
for $i \leftarrow 1$ to $N_{c'}$ **do**
  $\mathbf{S}_i = \frac{\mathbf{G}_i^{\mathbf{F}^{\mathcal{P}'}} (\mathbf{G}_i^{\mathbf{F}^{\mathcal{Q}'}})^T}{\tilde{d}}$
end for
$C_i \leftarrow$ select top K' in $\mathbf{S}_i$
$\mathcal{C}_{fine} \leftarrow C_1 \cup ... \cup C_{N_c}$
$\mathcal{T} \leftarrow$ Estimator($\mathcal{C}_{fine}$)
return $\mathcal{T}$

5 Experiments

Metrics. We use the following metrics to evaluate methods: Relative Rotation Error (**RRE**), Relative Translation Error (**RTE**), Registration Recall (**RR**) and Inlier Ratio (**IR**). For the Cross3DReg dataset, the threshold for RR is defined as RRE $< 2°$ and RTE < 0.5 m. The IR evaluates the quality of the point matching by calculating the proportion of corresponding points whose distances under the true transformation are below the threshold of 1.0m.

Implementation Details. All experiments are implemented based on the PyTorch framework and trained on the NVIDIA RTX A6000 GPU with the following key parameters: initial learning rate is 10^{-4}; Batch size is 1 and the weight decay is 10^{-6}. Our model is trained with Adam optimizer for 10 epochs. In addition, the core configurations of our model is the same in the KITTI-CrossSource and Cross3DReg.

The experiment setup on KITTI-CrossSource dataset. To isolate the challenges introduced by cross-sensor differences, similar to [9], we begin with synthetic experiments, where characteristics of different sensor setups are systematically modeled by sampling points from reconstructed point clouds. We apply random downampling, distance-based downsampling and farthest point sampling to the MonoRec-reconstructed point clouds, respectively. In our experiments, the source point clouds are the sampled MonoRec point clouds, while the target point clouds are the KITTI LiDAR point clouds. Specifically, the sampling ratio for random downsampling is set to 0.25 and the number of sampled points for farthest point sampling is set to $10,000$. Distance-based sampling is adopted to simulate the common spatial distribution of LiDAR point clouds, where nearby regions are densely sampled while distant regions are sparsely sampled. The core idea is that points farther away from the sensor origin have a lower probability of being retained. The retention probability is defined as: $p_i = p_{min} + (p_{max} - p_{min}) \times (1 - \bar{d}_i)^{\gamma}$, where $\bar{d}_i$ denotes the normalized distance within the range of $[0, 1]$, p_{max} controls the maximum retention probability for nearby points, p_{min} controls the minimum retention probability for distant points and γ controls the strength of distance decay. In our experiments, p_{max}, p_{min} and γ are set to 0.5, 0.05 and 4, respectively. It should be noted that all baseline methods are retrained for 20 epochs before evaluation. The registration recall (RR) is computed under the criterion that the relative rotation error is less than $5°$ and the relative translation error is less than 2m.

The experiment setup on Cross3DReg dataset. Due to the large number of points in outdoor scenes, we apply voxel downsampling with a voxel size of 0.25m to the input point clouds of all baseline methods for fair comparison. ICP [15] is implemented using Open3D with a point-to-point objective and the identity matrix is used as the initial transformation. The maximum correspondence distance is set to 1.5m and the maximum number of iterations is set to 200.

For learning-based methods, including FCGF [18], Omnet [25], Predator [13], CoFiNet [11], RoITr [10], IMFNet [44], GeoTrans [24], PARE-Net [12] and VRHCF [14], all approaches are trained on the Cross3DReg training set for 10 epochs and evaluated on the test set. FCGF [18] and IMFNet [44] are trained using SGD with a learning rate of 1×10^{-4} and a batch size of 1. The positive and negative distance thresholds in the contrastive loss are set to 0.1 and 1.4, respectively. Omnet [25] is

trained using Adam with a batch size of 1, a learning rate of 1×10^{-4} and $titer = 4$. Its loss parameters are set to $\alpha_1 = 1$, $\alpha_2 = 4$ and $\gamma = 1$. Predator is optimized using SGD with an initial learning rate of 0.05, a momentum of 0.98 and a weight decay of 1×10^{-6}. The learning rate scheduler with a decay factor of 0.99 per epoch is adopted. Its loss function consists of circle loss and overlap loss, both weighted by 1.0. The positive and negative margins for descriptor learning are set to 0.1 and 1.4, respectively. The RANSAC distance threshold during testing is set to 0.6m. CoFiNet is trained using Adam with an initial learning rate of 1×10^{-4} , a weight decay of 1×10^{-6} and the learning rate decay factor of 0.95. The batch size is set to 1. During matching, 128 candidate correspondences are selected, 64 neighboring points are used for each local patch and at least 200 coarse correspondences are retained. The matching loss and local matching loss are assigned equal weights. The RANSAC distance threshold is also set to 0.6m.

GeoTrans [24] employs a five-stage KPConv [38] backbone with an initial voxel size of 0.15 and a convolution radius of 4.25. The matching pipeline consists of coarse and fine matching stages. The coarse stage produces 256 superpoint correspondences, while the fine stage adopts mutual matching, top-2 selection strategy and five pose refinement steps. The model is trained via Adam optimizer with an initial learning rate of 1×10^{-4}, a batch size of 1 and a learning rate decay factor of 0.95 every 4 epochs. RoITr [10] is optimized by SGD optimizer with an initial learning rate of 0.05 and a convolution radius of 4.25. Its core parameter configurations for coarse and fine registration are consistent with those of GeoTrans and equal weights are assigned to the circle loss and overlap loss. PARE-Net [12] is trained using Adam with an initial learning rate of 1×10^{-4}, a batch size of 1 and a learning rate decay factor of 0.95 every 4 epochs. Its feature extraction network contains four sampling layers, where each layer leverages 35 neighbor points and adopts an initial voxel size of 0.3m. The coarse registration settings are also kept consistent with GeoTrans. In the fine registration stage, the acceptance radius is set to 0.6m and the number of hypotheses is set to 1000. The total loss function is composed of coarse-level matching loss, fine-level rotation-invariant feature loss and fine-level rotation-equivariant feature loss, all of which are weighted equally.

5.1 Quantitative Comparison

Evaluation on KITTI-CrossSource. The experimental result is shown in Table 2. Overall, different sampling strategies simulate characteristics of different sensor setups. The conventional methods RoITr [10] and PARE-Net[12] exhibit large performance fluctuations under the three sampling settings. In particular, under distance-based sampling, RoITr [10] obtains RRE and RTE values of 54.52 and 18.87, respectively, while its RR drops to 18.2%. This suggests that non-uniform point distributions, with dense nearby measurements and sparse distant measurements, can substantially affect feature matching by introducing density-induced ambiguities, thereby increasing incorrect correspondences and degrading pose estimation accuracy. VRHCF [14] achieves an RR of 0.0% under all sampling settings, suggesting that it struggles to obtain stable and effective registration results in the current cross-source sensor simulation scenarios. In contrast, GeoTrans [24] achieves relatively high registration recall

Table 2 Registration results under different sampling strategies on KITTI-CrossSource.

Method	Random Sampling			Distance-based Sampling			Farthest Point Sampling		
	RRE(°) ↓	RTE(m) ↓	RR(%) ↑	RRE(°) ↓	RTE(m) ↓	RR(%) ↑	RRE(°) ↓	RTE(m) ↓	RR(%) ↑
RoITr [10]	42.28	14.87	26.3%	54.52	18.87	18.2%	40.15	15.51	26.8%
PARE-Net [12]	15.78	6.25	22.6%	15.69	5.09	20.8%	20.01	7.06	20.0%
GeoTrans [24]	**1.77**	1.10	91.2%	**1.79**	**0.74**	90.8%	2.12	1.18	89.7%
VRHCF [14]	28.80	41.39	0.0%	12.24	32.22	0.0%	10.14	15.57	0.0%
Ours	2.09	**0.88**	**94.3%**	1.90	0.88	**92.2%**	**1.46**	**0.69**	**95.6%**

under all three sampling strategies, demonstrating strong baseline performance for cross-source point cloud registration. Under random sampling and distance-based sampling, GeoTrans achieves RR values of 91.2% and 90.8%, respectively and obtains the lowest RTE of 0.74 under distance-based sampling. However, under farthest point sampling, its RR decreases to 89.7%, while its RRE and RTE increase to 2.12 and 1.18, respectively. This suggests that when the sampling process tends to preserve spatially dispersed points, the local geometric structure and density distribution are altered, which still affects the stability of feature matching. Notably, both PARE-Net and GeoTrans perform worse with farthest point sampling (FPS) than with random sampling. We attribute this to their sensitivity to local perturbations: without denoising, FPS is more likely to retain spatially isolated noise points. Since both methods rely on local-neighborhood features and geometric matching, these retained outliers weaken feature consistency and thus degrade registration performance. This also explains why RoITr and our method are more robust: by fusing hand-crafted PPF features and image features, respectively, they introduce consistent cross-modal cues that counteract such local perturbations.

Our method achieves more stable overall performance under all three sampling settings. Under random sampling, our method obtains the highest registration recall, with an RR of 94.3% and achieves the lowest translation error of 0.88. Under distance-based sampling, our method also achieves the highest RR of 92.2%, indicating that it can better adapt to the non-uniform point density distribution caused by distance-dependent attenuation. Under farthest point sampling, our method further achieves the best RRE, RTE and RR, reaching 1.46, 0.69 and 95.6%, respectively. These results demonstrate that even when different sampling strategies lead to significant changes in local point density, spatial coverage and structural distribution, the proposed method can still extract more stable and discriminative cross-source features, thereby enabling more accurate and robust pose estimation.

Evaluation on Cross3DReg. To validate the effectiveness of the proposed method and evaluate its registration performance on the real-world scans, we conduct a comparative experiment with several state-of-the-art point cloud registration approaches on Cross3DReg dataset, as summarized in Table 3. The selected methods cover a diverse spectrum of methodologies, including a traditional iterative optimization method (ICP [15]); correspondence-based approaches (FCGF [18], Omnet [25], Predator [13], CoFiNet [11], RoITr [10], GeoTrans [24], PARE-Net [12]). In addition, we compare with VRHCF [14], GCTR [32] and cross-PCR [1], as well as the multimodal registration method IMFNet [44]. The results show that our approach significantly outperforms the state-of-the-art methods in all evaluation metrics. In terms

of registration accuracy, the lowest RRE = 6.68° is achieved. Our method also achieves the RR metrics, which is 4.5% higher than the state-of-the-art GeoTrans.

Table 3 The comparison of the same- and cross-source registration methods on Cross3DReg. Method marked with * is our own reimplementation, not the official code released by the original authors.

	Methods	RRE(°) ↓	RTE(m)↓	RR(%) ↑
Same-source	ICP [15]	93.61	9.72	1.8
	FCGF [18]	94.70	9.00	0.0
	Omnet [25]	95.01	11.20	0.0
	Predator [13]	106.58	32.08	0.0
	CoFiNet [11]	125.62	62.07	0.0
	RoITr [10]	16.76	2.95	53.9
	IMFNet [44]	94.8	9.07	0.0
	GeoTrans [24]	18.15	1.69	81.7
	PARE-Net [12]	6.69	0.96	68.4
Cross-source	VRHCF [14]	110.62	16.35	0.0
	GCTR* [32]	111.10	20.37	0.0
	Cross-PCR* [1]	49.83	17.17	15.1
	Cross3DReg (Ours)	**6.68**	1.01	**86.2**

It should be noted that the RR measures the proportion of test pairs for which both the relative rotation error and the relative translation error are below the specified thresholds. In Table 3, GCTR and VRHCF obtain registration recalls of 0.0%, with RRE values of 111.10° and 110.62°, respectively. These results indicate that their estimated transformations rarely satisfy the joint rotation and translation criteria. GCTR relies on regional affinity and point-wise refinement, whereas VRHCF uses voxel-based representation and hierarchical correspondence filtering. The non-uniform density, sensor-specific scanning patterns, outliers and missing regions in the two physical LiDAR scans can weaken the reliability of these representations and lead to incorrect correspondences. FCGF and Predator rely heavily on the consistency of learned local geometric descriptors, while CoFiNet depends on reliable coarse-level superpoint correspondences before fine matching. The density variations, ring-shaped versus non-repetitive scanning patterns, noise, outliers and low-overlap regions in Cross3DReg violate these assumptions and produce many incorrect correspondences. Consequently, the subsequent pose estimation stage cannot reliably identify a consensus transformation. For Omnet, the learned overlap prediction is also affected by the geometric discrepancy between the two sources. IMFNet further suffers from global multimodal fusion in the uncalibrated camera–LiDAR setting, where non-corresponding image features can be introduced into the point-cloud representation. GeoTrans, RoITr, Cross-PCR and PARE-Net mitigate this issue by employing geometrically invariant or rotation-aware representations together with coarse-to-fine matching, yet their coarse correspondences still contain varying degrees of outliers.

Our method additionally filters irrelevant points with the Overlapping Mask Predictor and uses shared image context to improve feature consistency. This explains its higher recall and is also consistent with the correspondence visualization in Fig 7.

Table 4 Registration performance on the Cross3DReg dataset with different pose estimators.

						Runtime (s)↓		
Estimator	Method	RRE(°) ↓	RTE(m)↓	RR(%) ↑	IR(%) ↑	model inference	pose estimation	Total
LGR	Geotrans [24]	18.148	1.690	81.7	60.1	**0.186**	0.032	**0.218**
	RoITr [10]	16.718	3.098	53.1	25.8	0.245	0.033	0.278
	Ours	**6.683**	**1.010**	**87.1**	**70.3**	0.252	**0.031**	0.283
RANSAC-50k	RoITr [10]	9.731	2.923	67.8	25.8	0.245	0.144	0.389
	Predator [13]	100.81	30.37	0.0	0.25	0.354	0.564	0.918
	CoFiNet [11]	99.34	16.97	0.0	0.13	**0.073**	**0.026**	**0.099**
	Geotrans [24]	13.05	1.436	72.2	60.1	0.186	1.299	1.485
	Ours	**8.150**	**1.135**	**78.6**	**70.3**	0.252	1.353	1.605
weighted SVD	Geotrans [24]	16.491	2.193	**62.9**	60.1	**0.186**	**0.002**	**0.188**
	RoITr [10]	54.21	16.54	0.2	25.8	0.245	0.003	0.248
	Ours	**8.684**	**1.153**	57.8	**70.3**	0.252	0.002	0.254

To fully evaluate the correspondence of our cross-source point cloud registration method, we evaluate the registration accuracy with different estimators. As shown in Table 4, when combined with the LGR [24] pose estimator, our method exhibits optimal performance in all key evaluation metrics. When we employ the RANSAC estimator, our method still outperforms other methods, with RRE = 8.150° and RTE = 1.135 m, significantly better than GeoTrans, Predator and CoFiNet, which also employ RANSAC. It is worth noting that both Predator and CoFiNet have an RR of 0.0% in this setup, indicating that they fail to produce correspondences of sufficient quality for the RANSAC estimator to achieve accurate alignment. Finally, we perform the tests with using the weighted SVD [15], our method still achieves accurate registration where RRE=8.684°, RTE=1.153 m and registration recall reaches 57.8%. In contrast, the registration recall of RoITr is only 0.2%. Furthermore, it is observed that despite our method achieving approximately 10% higher inlier ratio and lower registration error compared to GeoTrans, there is a gap in registration recall. The core reason for this is not the low quality of our generated correspondences, but rather the insufficient robustness of the pose estimator. Specifically, weighted SVD, a least-squares solution, attempts to fit all given corresponding points. This makes it highly sensitive to the mismatched correspondences. Moreover, we also evaluate the computational efficiency of our approach. We categorize the runtime into model inference, pose estimation and total inference time. Compared with GeoTrans, the runtime of our method increases by 29.8%, in which the model inference increases by 35.5%. This is because our method utilizes image features to enhance the cross-source point cloud feature consistency. Benefiting from this design, the RRE and RTE decrease by 63.2% and 40.2%, respectively. The runtime of pose estimation is comparable with other methods.

5.2 Qualitative Comparisons.

Besides the quantitative comparison, in Figure 6, we also present a qualitative comparison of our method against current baseline approaches in the selected three challenging scenes, which has significant differences in density and pattern. As can be observed, methods such as Predator and CoFiNet fail to achieve successful alignment in the three challenging scenarios. In contrast to RoITr and GeoTrans, our method attains the optimal registration performance, with results that are visually closest to the Ground Truth.

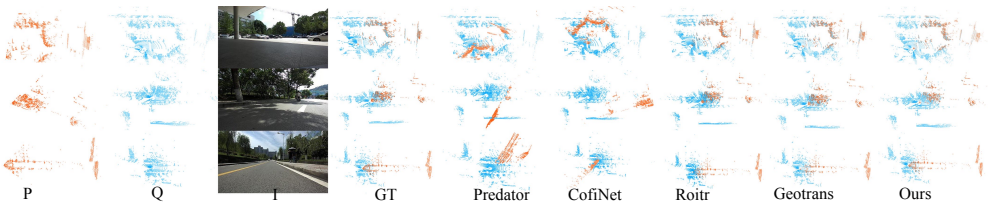


Fig. 6 The qualitative comparison on Cross3DReg.

Figure 7 visualizes the point correspondences generated by different registration approaches. The qualitative comparison clearly reveals that, compared to current state-of-the-art methods, the correspondences extracted by our method are more precise and highly concentrated on the overlapping regions of the point clouds. In contrast, methods like Predator, CoFiNet and RoITr produce a substantial number of incorrect correspondences. We attribute this primarily to two factors: a significant decrease in feature consistency within the overlapping regions when faced with considerable density variations across point clouds and severe interference in feature matching caused by prominent noise in the scenes.

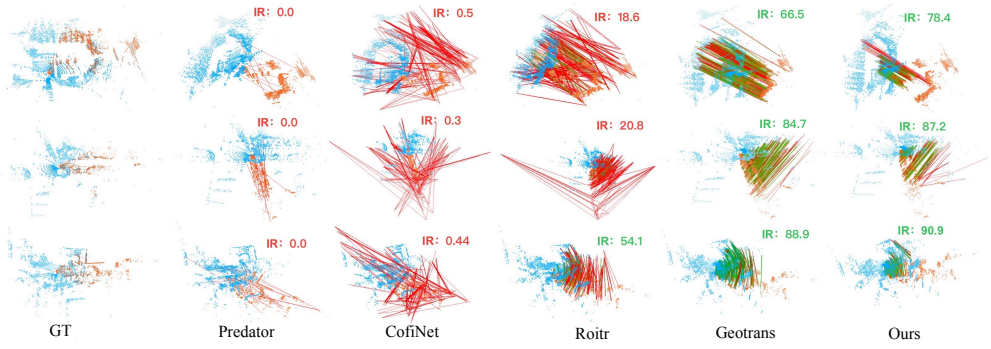


Fig. 7 The visualization of correspondences on Cross3DReg.

To further analyze the role of 2D image features in improving the consistency of cross-source point cloud feature between the source and target point clouds, Figure

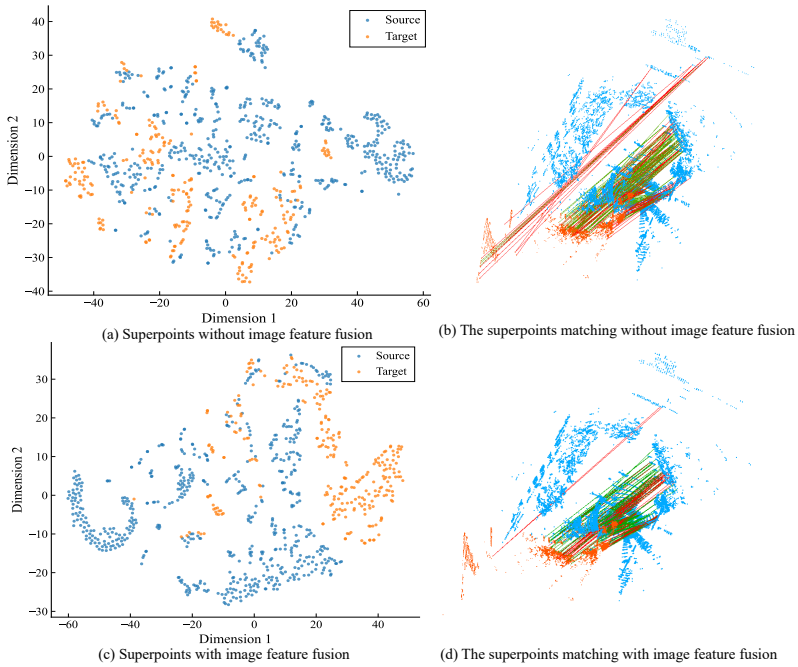


Fig. 8 Comparison of cross-source point cloud t-SNE distributions and superpoints matching with and without image feature fusion.

8 compares the visualization of t-SNE feature distributions and superpoint matching results with and without image feature fusion. Without image feature fusion, the superpoint features of the source and target point clouds are highly mixed in the feature space, making it difficult to form discriminative cross-source feature representations. As a result, the generated correspondences contain a large number of ambiguous and unreliable matches. In contrast, after introducing image feature fusion, the superpoint features of the source and target point clouds exhibit a more consistent structural distribution and show stronger overlap relationships in local regions. Meanwhile, the corresponding superpoint matches are more accurate and demonstrate better spatial consistency. These results indicate that 2D image features can provide complementary contextual information for cross-source point clouds. By introducing image-guided information, the proposed method effectively alleviates the feature inconsistency caused by sensor discrepancies and improves the reliability and spatial consistency of superpoint matching.

Additionally, as shown in Figure 9, we provide a visual comparison of the overlapping regions between our proposed method and Omnet to show the necessity of using a common image to predict the overlap region. It can be observed that the point cloud overlap region estimated by our method aligns more closely with the ground truth, whereas Omnet's results exhibit noticeable deviations. This discrepancy arises because Omnet directly regresses the overlapping regions from the features of two

point cloud frames. In cross-source outdoor scenarios, however, point clouds are typically structurally different and of different levels of noise and outliers, leading to low feature similarity between point clouds. As a result, the model struggles to accurately locate overlap regions, only relying on the geometric features. Incorrect predictions of overlap regions subsequently cause incorrect matches, resulting in registration failure.

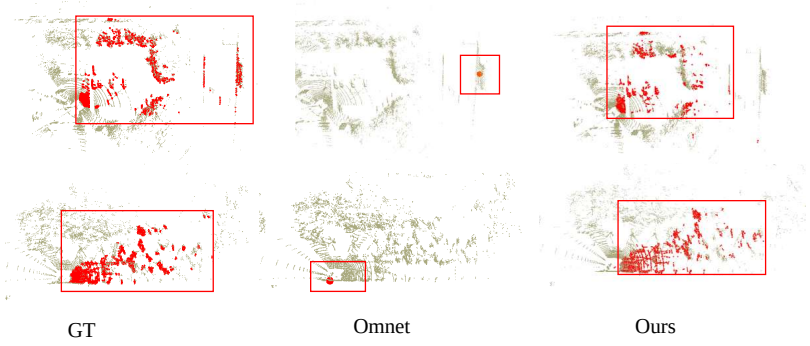


Fig. 9 The comparison of overlap regions extracted with our method and Omnet.

Figure 10 shows the OMP module prediction results. Given the unaligned image $\mathbf{I}$, source and target point clouds $\mathcal{P}$ and $\mathcal{Q}$, we visualize the predicted masked point cloud area, which is the highlighted parts. We also show the visual comparison of the registration results between the proposed Cross3DReg method and ground truth.

To evaluate the robustness and generalization of the proposed method, we firstly evaluate the registration accuracy when different image sequences are adopted as visual information. As illustrated in Figure 11, we select the original input image alongside frames positioned 20 frames before and after the current frame for comparison. The results demonstrate that our approach maintains stability even when confronted with input images captured from other viewpoints.

We also evaluate the scenarios where images are captured with different installation positions of RGB cameras. Here, we test the registration results when the visual

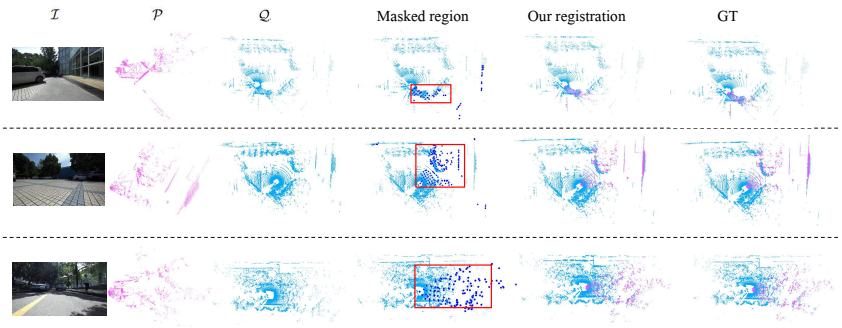


Fig. 10 Additional visualization of Cross3DReg registration results.

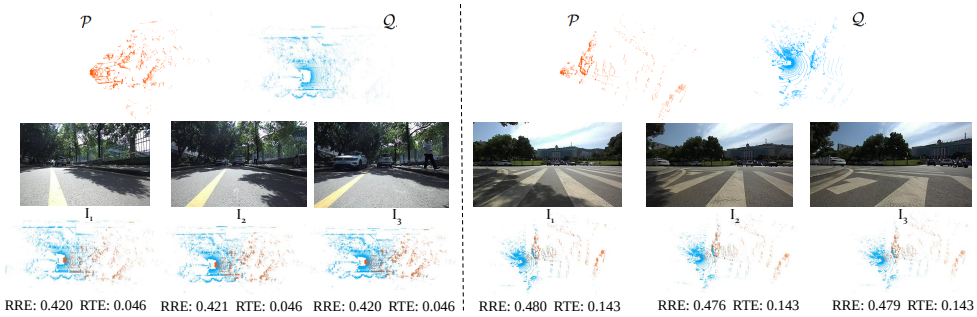


Fig. 11 The registration when input images are captured from other views. $\mathbf{I}_1$ represents the image 20 frames prior to $\mathbf{I}_2$, while $\mathbf{I}_3$ denotes the image 20 frames subsequent to $\mathbf{I}_2$.

information is obtained with images of the top RGB camera on our autonomous vehicle platform. As we can see in Figure 12, $\mathbf{I}_1$ is the original image from ZED camera and $\mathbf{I}_2$ is the image from the top RGB camera. Images from the top RGB camera present obvious distortion and some examples are overexposed where details cannot be seen. With these challenging and imperfect images, our approach still maintains superior performance in such cases. These experiments verify that our method is robust to different hardware installation settings, low-quality images and is easy to generalize to arbitrary camera and LiDAR relative positions.

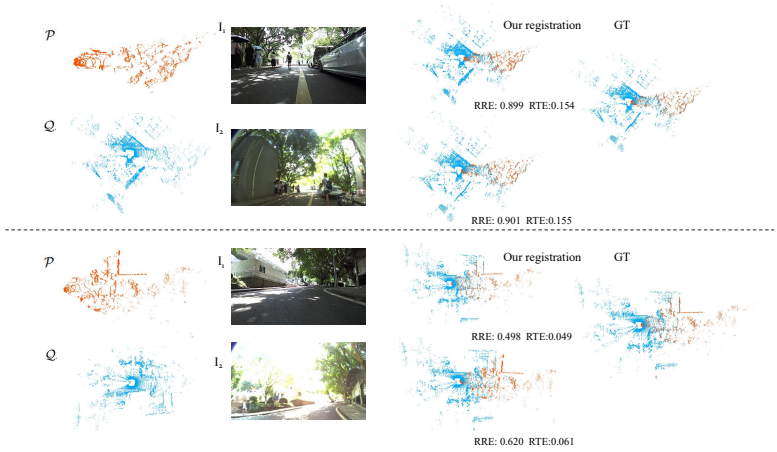
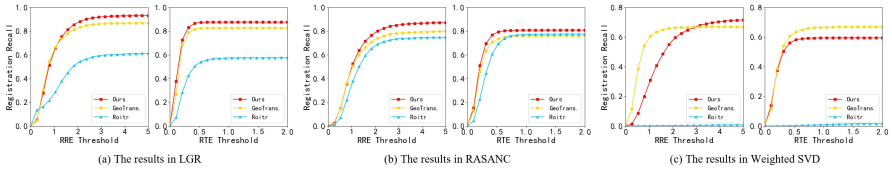


Fig. 12 Visualized registration results using a distorted and overexposed image captured from the RGB camera installed at a different position.

Furthermore, Figure 13 displays the registration recall of each method under different relative rotation and translation error thresholds, utilizing different pose estimators. The results demonstrate that when robust pose estimators such as LGR and

RANSAC are employed, the poses computed from the correspondences generated by our method exhibit the highest registration recall across all error thresholds.



(a) The results in LGR (b) The results in RASANC (c) The results in Weighted SVD

Fig. 13 Registration recalls with different RRE and RTE thresholds in different estimators on Cross3DReg.

5.3 Ablation Studies

As shown in Table 5, we conduct ablation experiments on the Cross3DReg dataset to evaluate the effectiveness of modules proposed in the method. Experiments are set up

Table 5 The ablation study of each module of the proposed method.

Method	RRE(°)↓	RTE(m)↓	RR(%)↑	IR(%)↑
(a) Geo self-attention w/o OMP	18.184	1.690	81.7	60.1
(b) VGAM w/o OMP	17.812	1.592	82.1	60.3
(c) OMP w/ Vanilla self-attention	8.496	1.242	85.3	70.0
(d) OMP w/ Geo self-attention	8.371	1.287	86.7	70.1
(e) OMP w/ VGAM(full)	**6.683**	**1.010**	**87.2**	**70.3**

with five scenarios in comparison: (a) only the geometric self-attention module is used; (b) only the VGAM is used; (c) the OMP module is used with the vanilla attention module; (d) the OMP module is used with the geometric self-attention module; and (e) the complete Cross3DReg method. The comparison among (a), (b), (e) shows the OMP module can effectively mitigate the interference of redundant and noisy points, significantly improving the accuracy of cross-source point cloud alignment. The comparison of schemes (b), (c), (d) and (e) further shows that our visual-geometric feature attention guidance module improves the consistency of feature space among cross-source point clouds, enhancing the registration accuracy.

6 Conclusion

In this paper, we propose a real-world cross-source point cloud registration dataset, named Cross3DReg, to show different levels of noise, outliers, densities and structural patterns. An overlap-based cross-source point cloud registration method is then proposed to achieve accurate registration by predicting the overlap region with the help of unaligned images and ignoring redundant points and noise. Extensive experiments verify the challenges of our proposed dataset and the effectiveness of the proposed method.

7 Declarations

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Data Availability. Our Cross3DReg dataset will be released on GitHub at https://github.com/bestmyselflllll/Cross3DReg. Data from 3DCSR and KITTI-CrossSource can also be downloaded from their respective repositories.

Conflict interest. The authors declare that they have no conflict of interest.

References

[1] Zhao, G., Guo, Z., Du, Z., Ma, H.: Cross-pcr: A robust cross-source point cloud registration framework. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, pp. 10403–10411 (2025)

[2] Huang, X., Mei, G., Zhang, J., Abbas, R.: A comprehensive survey on point cloud registration. arXiv preprint arXiv:2103.02690 (2021)

[3] Huang, X., Mei, G., Zhang, J.: Cross-source point cloud registration: Challenges, progress and prospects. Neurocomputing **548**, 126383 (2023)

[4] Yin, H., Xu, X., Lu, S., Chen, X., Xiong, R., Shen, S., Stachniss, C., Wang, Y.: A survey on global lidar localization: Challenges, advances and open problems. International Journal of Computer Vision **132**(8), 3139–3171 (2024)

[5] Roldao, L., De Charette, R., Verroust-Blondet, A.: 3d semantic scene completion: A survey. International Journal of Computer Vision **130**(8), 1978–2005 (2022)

[6] Zhu, Z., Chen, H., He, X., Wei, M.: Pointsea: Point cloud completion via self-structure augmentation. International Journal of Computer Vision **133**(7), 4770–4794 (2025)

[7] Huang, X., Mei, G., Zhang, J.: Cross-source point cloud registration: Challenges, progress and prospects. Neurocomputing **548**, 126383 (2023)

[8] Xiong, K., Zheng, M., Xu, Q., Wen, C., Shen, S., Wang, C.: Speal: Skeletal prior embedded attention learning for cross-source point cloud registration. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, pp. 6279–6287 (2024)

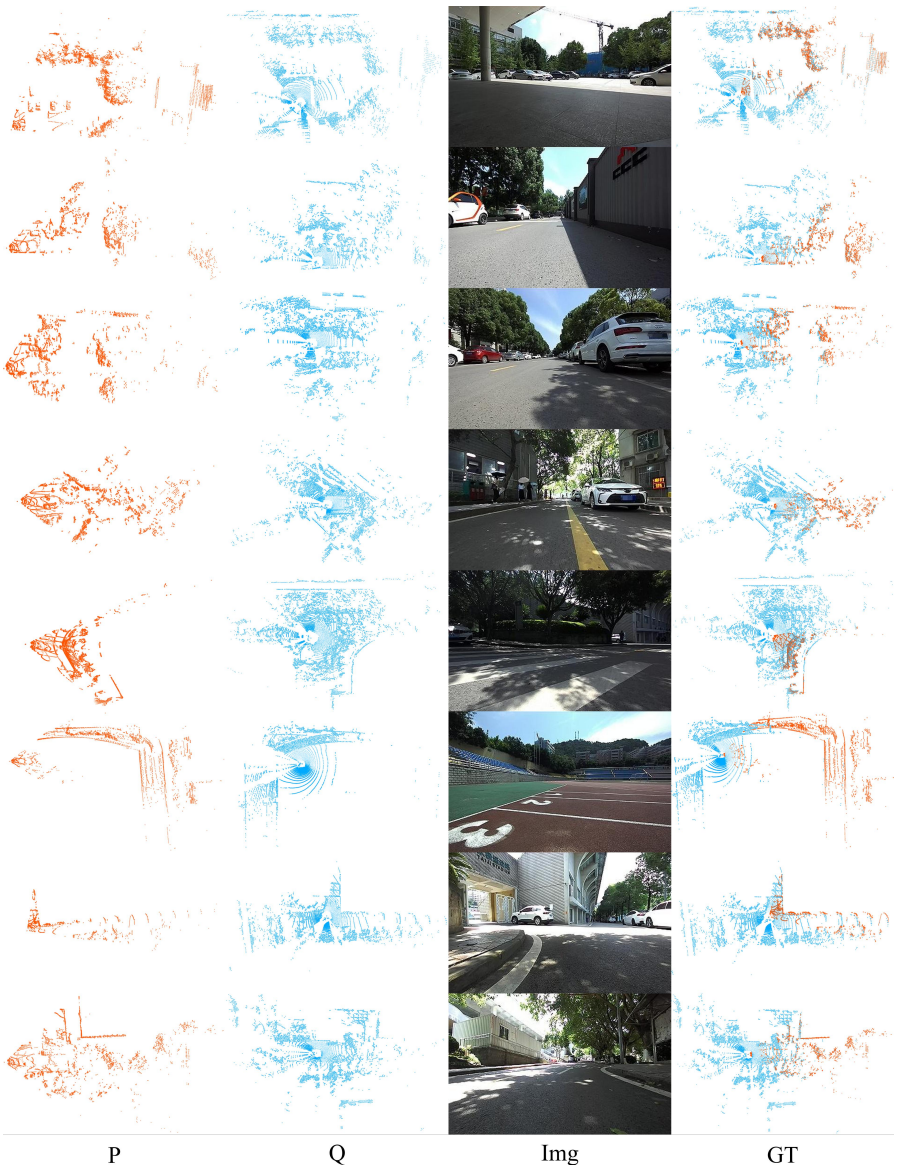


Fig. 14 The visualization of Cross3DReg dataset.

[9] Groß, J., Ošep, A., Leibe, B.: Alignnet-3d: Fast point cloud registration of partially observed objects. In: 2019 International Conference on 3d Vision (3DV), pp. 623–632 (2019). IEEE

[10] Yu, H., Qin, Z., Hou, J., Saleh, M., Li, D., Busam, B., Ilic, S.: Rotation-invariant transformer for point cloud matching. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5384–5393 (2023)

[11] Yu, H., Li, F., Saleh, M., Busam, B., Ilic, S.: Cofinet: Reliable coarse-to-fine correspondences for robust pointcloud registration. Advances in Neural Information Processing Systems **34**, 23872–23884 (2021)

[12] Yao, R., Du, S., Cui, W., Tang, C., Yang, C.: Pare-net: Position-aware rotation-equivariant networks for robust point cloud registration. In: European Conference on Computer Vision, pp. 287–303 (2024). Springer

[13] Huang, S., Gojcic, Z., Usvyatsov, M., Wieser, A., Schindler, K.: Predator: Registration of 3d point clouds with low overlap. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 4267–4276 (2021)

[14] Zhao, G., Du, Z., Guo, Z., Ma, H.: Vrhcf: Cross-source point cloud registration via voxel representation and hierarchical correspondence filtering. In: 2024 IEEE International Conference on Multimedia and Expo (ICME), pp. 1–6 (2024). IEEE

[15] Besl, P.J., McKay, N.D.: Method for registration of 3-d shapes. In: Sensor Fusion IV: Control Paradigms and Data Structures, vol. 1611, pp. 586–606 (1992). Spie

[16] Rusinkiewicz, S., Levoy, M.: Efficient variants of the icp algorithm. In: Proceedings Third International Conference on 3-D Digital Imaging and Modeling, pp. 145–152 (2001). IEEE

[17] Segal, A., Haehnel, D., Thrun, S.: Generalized-icp. In: Robotics: Science and Systems, vol. 2, p. 435 (2009). Seattle, WA

[18] Choy, C., Park, J., Koltun, V.: Fully convolutional geometric features. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 8958–8966 (2019)

[19] Wang, H., Liu, Y., Dong, Z., Wang, W.: You only hypothesize once: Point cloud registration with rotation-equivariant descriptors. In: Proceedings of the 30th ACM International Conference on Multimedia, pp. 1630–1641 (2022)

[20] Rusu, R.B., Blodow, N., Marton, Z.C., Beetz, M.: Aligning point cloud views using persistent feature histograms. In: 2008 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 3384–3391 (2008). IEEE

[21] Rusu, R.B., Blodow, N., Beetz, M.: Fast point feature histograms (fpfh) for 3d registration. In: 2009 IEEE International Conference on Robotics and Automation, pp. 3212–3217 (2009). IEEE

[22] Bai, X., Luo, Z., Zhou, L., Fu, H., Quan, L., Tai, C.-L.: D3feat: Joint learning of dense detection and description of 3d local features. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 6359–6367 (2020)

[23] Ao, S., Hu, Q., Yang, B., Markham, A., Guo, Y.: Spinnet: Learning a general surface descriptor for 3d point cloud registration. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 11753–11762 (2021)

[24] Qin, Z., Yu, H., Wang, C., Guo, Y., Peng, Y., Ilic, S., Hu, D., Xu, K.: Geotransformer: Fast and robust point cloud registration with geometric transformer. IEEE Transactions on Pattern Analysis and Machine Intelligence **45**(8), 9806–9821 (2023)

[25] Xu, H., Liu, S., Wang, G., Liu, G., Zeng, B.: Omnet: Learning overlapping mask for partial-to-partial point cloud registration. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 3132–3141 (2021)

[26] Zhang, Z., Sun, J., Dai, Y., Zhou, D., Song, X., He, M.: End-to-end learning the partial permutation matrix for robust 3d point cloud registration. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 36, pp. 3399–3407 (2022)

[27] Lu, W., Wan, G., Zhou, Y., Fu, X., Yuan, P., Song, S.: Deepvcp: An end-to-end deep neural network for point cloud registration. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 12–21 (2019)

[28] Zhang, Y., Yu, J., Huang, X., Zhou, W., Hou, J.: Pcr-cg: Point cloud registration via deep explicit color and geometry. In: European Conference on Computer Vision, pp. 443–459 (2022). Springer

[29] Xu, Z., Jiang, X., Gao, X., Gao, R., Gu, C., Zhang, Q., Li, W., Gao, X.: Igreg: Image-geometry-assisted point cloud registration via selective correlation fusion. IEEE Transactions on Multimedia (2024)

[30] Xu, Z., Gao, X., Jiang, X., Cheng, S., Zhang, Q., Li, W., Gao, X.: S2reg: Structure-semantics collaborative point cloud registration. Pattern Recognition **161**, 111290 (2025)

[31] Huang, X., Zhang, J., Wu, Q., Fan, L., Yuan, C.: A coarse-to-fine algorithm for matching and registration in 3d cross-source point clouds. IEEE Transactions on Circuits and Systems for Video Technology **28**(10), 2965–2977 (2017)

[32] Huang, X., Fan, L., Wu, Q., Zhang, J., Yuan, C.: Fast registration for cross-source point clouds by using weak regional affinity and pixel-wise refinement. In: 2019 IEEE International Conference on Multimedia and Expo (ICME), pp. 1552–1557 (2019). IEEE

[33] Ma, N., Wang, M., Han, Y., Liu, Y.-J.: Ff-logo: Cross-modality point cloud registration with feature filtering and local to global optimization. In: 2024 IEEE International Conference on Robotics and Automation (ICRA), pp. 744–750 (2024). IEEE

[34] Zeng, A., Song, S., Nießner, M., Fisher, M., Xiao, J., Funkhouser, T.: 3dmatch: Learning local geometric descriptors from rgb-d reconstructions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1802–1811 (2017)

[35] Geiger, A., Lenz, P., Stiller, C., Urtasun, R.: Vision meets robotics: The kitti dataset. The international journal of robotics research **32**(11), 1231–1237 (2013)

[36] Liang, J., Deng, Z., Zhou, Z., Sun, M., Ghasemalizadeh, O., Kuo, C.-H., Sen, A., Manocha, D.: Cscpr: Cross-source-context indoor rgb-d place recognition. IEEE Robotics and Automation Letters (2025)

[37] Chen, Z., Qi, Y., Feng, D., Zhuang, X., Chen, H., Hu, X., Wu, J., Peng, K., Lu, P.: Heterogeneous lidar dataset for benchmarking robust localization in diverse degenerate scenarios. The International Journal of Robotics Research, 02783649251344967 (2024)

[38] Thomas, H., Qi, C.R., Deschaud, J.-E., Marcotegui, B., Goulette, F., Guibas, L.J.: Kpconv: Flexible and deformable convolution for point clouds. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 6411–6420 (2019)

[39] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. Advances in neural information processing systems **30** (2017)

[40] Rocco, I., Cimpoi, M., Arandjelović, R., Torii, A., Pajdla, T., Sivic, J.: Neighbourhood consensus networks. Advances in neural information processing systems **31** (2018)

[41] Sun, J., Shen, Z., Wang, Y., Bao, H., Zhou, X.: Loftr: Detector-free local feature matching with transformers. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 8922–8931 (2021)

[42] Sarlin, P.-E., DeTone, D., Malisiewicz, T., Rabinovich, A.: Superglue: Learning feature matching with graph neural networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 4938–4947 (2020)

[43] Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2980–2988 (2017)

[44] Huang, X., Qu, W., Zuo, Y., Fang, Y., Zhao, X.: Imfnet: Interpretable multimodal fusion for point cloud registration. IEEE Robotics and Automation Letters **7**(4), 12323–12330 (2022)